Safety-Aware Vehicle Routing for Quick Commerce Delivery: A Behavioral Multi-Objective Optimization Approach

Research Proposal

March 6, 2026

Abstract

Quick commerce platforms have transformed urban logistics by delivering goods within tight time windows, typically ten to thirty minutes. This heavy emphasis on speed requires aggressive routing strategies that often risk road safety and put immense pressure on delivery riders. To address this issue, this paper proposes the Safety-Aware Vehicle Routing Problem with Time Windows (SA-VRPTW). This framework integrates real-time risk measurements, infrastructure data, and rider behavior into the routing model. Operating as a true multi-objective Pareto optimization challenge, the system balances probabilistic collision risk against expected delivery time. Crucially, it replaces arbitrary hard deadlines with Soft Time Windows (STW) yielding exponential financial lateness penalties, and introduces hard constraints for cumulative rider behavioral stress and road-hierarchy protections to prevent shifting hazards to vulnerable neighborhoods. To implement this framework, we design a data pipeline using OpenStreetMap for road networks and the Integrated Road Accident Database (iRAD) for crash statistics. A critical requirement for this system is obtaining authorized access to governmental iRAD records to validate the risk models. Finally, we evaluate exact, metaheuristic, and learning-based solution methods for this computationally demanding problem. This research aims to show that rider safety can be a primary goal in routing, leading to safer and more sustainable urban logistics.

1 Introduction

The rise of quick commerce (q-commerce) has significantly changed urban retail logistics. By focusing on rapid, hyper-local delivery speed rather than just product variety, platforms now promise deliveries in mere minutes. This model relies on micro-fulfillment centers, automated real-time dispatch systems, and a large fleet of gig-economy delivery riders. While highly convenient for consumers, this relentless focus on speed creates serious issues. The most notable problems are the increased risk of traffic accidents and the physical and psychological stress placed on riders.

Historically, the Vehicle Routing Problem (VRP) has mainly focused on minimizing costs, such as reducing the total distance or fleet size. However, the modern urban environment requires a broader perspective. The routing challenge has become a multi-objective optimization problem that must include safety, real-time risk, and human behavior. In q-commerce, delivery partners often work under incentive systems tied to strict completion times and daily targets. As deadlines approach, riders are encouraged to choose shorter but riskier routes, violate traffic rules, or speed up. In this way, algorithms directly increase systemic safety risks in busy cities.

To solve this problem, we propose the Safety-Aware Vehicle Routing Problem with Time Windows (SA-VRPTW) framework. This model establishes a multi-objective Pareto optimization framework prioritizing deterministic pathfinding. Recognizing the inherent unit mismatches in naive summation models, we formulate safety mathematically as route survival probability rather than additive penalties. Furthermore, we expand the scope of risk data beyond historical crashes to incorporate Surrogate Safety Measures (SSM) and dynamic weather-state multipliers. We propose combining static road data, real-time traffic updates, and historical accident records. To do this, our framework relies on the government-maintained Integrated Road Accident Database (iRAD). Using this data requires securing formal access rights from municipal or state authorities. By adding real crash data to the city map, our system balances delivery efficiency with congestion and collision risk, prioritizing public safety.

Building a truly safety-aware system requires reliable, high-quality data. Utilizing historical accident locations, severities, and times is essential. This makes accessing official traffic databases a core part of our methodology. At the same time, we must evaluate different solution techniques, from exact mathematical solvers to advanced metaheuristics, to ensure the complex safety calculations do not severely slow down the real-time routing needed for operations.

2 Literature Review

This research builds on classical optimization, risk assessment, and the new field of behavioral logistics. Un-

derstanding this landscape requires combining these traditionally separate areas.

2.1 Vehicle Routing Problem with Time Windows

The mathematical basis for q-commerce logistics is the Vehicle Routing Problem with Time Windows (VRPTW). This extends the standard VRP by adding strict delivery time intervals. Solomon (1987) created the early heuristic algorithms and standard benchmarks for this problem. Later, Desrochers et al. (1992) introduced exact solution methods using column generation, proving that optimal solutions were achievable for smaller, constrained problems.

However, modern q-commerce scale presents huge challenges. High order volumes, random demand, and the need for fast dispatching make exact methods too slow for real-world use. The VRPTW is fundamentally an NP-hard problem. Because of this, modern systems rely on metaheuristics like Genetic Algorithms and Tabu Search, which quickly find good, if not perfect, routes. Recently, Deep Reinforcement Learning has gained attention for its ability to learn routing policies from data and adjust on the fly.

2.2 Safety and Risk Quantification

Adding safety metrics to routing first started in Hazardous Materials (Hazmat) transportation. Early models, like those by Zografos and Androutsopoulos (2004), focused on minimizing the exposure of large populations to dangerous materials.

Recently, researchers have applied these ideas to everyday urban logistics. Hoseinzadeh et al. (2020) created a 'Safety Index' that uses historical crash data and real-time driver behavior telematics to score road danger. Building on this, the Risk-Aware Dynamic Routing (RADR) framework uses advanced graph learning to avoid high-risk road segments. These methods have shown that delivery fleets can reduce accident risk by over 18% with only a minor impact on overall travel times.

2.3 Behavioral Logistics

Even with good algorithms, safety-aware routing depends on the human riders. Behavioral logistics studies how algorithmic instructions affect human choices. The gig economy creates specific behavioral barriers to safety.

Sinchaisri et al. (2023) showed that gig riders often follow non-rational patterns like 'income-targeting,' where financial goals dictate their risk tolerance. Salmon et al. (2023) argued that traffic accidents in q-commerce are not just human errors, but 'systemic failures'. The algorithm's pressure for extreme speed structurally forces riders to take risks.

Fixing this requires smart design. Studies show that 'information nudging'—giving riders real-time data about route risks—along with financial incentives for safety, can change behavior. By unlinking pay from pure speed and rewarding safe driving, platforms can significantly improve fleet safety.

2.4 Addressing Research Gaps

While research on optimization, risk modeling, and gig-economy behavior exists, there is a lack of integrated studies. Specifically, little research explores the engineering scale needed to apply national crash databases to dispatching algorithms. Also, behavioral models rarely track rider fatigue over long shifts. This proposal addresses these gaps by creating a unified mathematical framework that balances the commercial need for fast delivery with the ethical need for safety.

3 System Architecture and Conceptual Framework

Creating a safety-aware routing ecosystem requires a two-layer architecture. The macro-layer handles large environmental datasets, while the micro-layer addresses the delivery partner's behavior.

As shown in Figure 1, the main routing engine uses several data streams. The road layout comes from OpenStreetMap, determining basic travel times. At the same time, the government's iRAD database provides crash history, setting the risk levels for each road. Live traffic APIs, like Google Maps, add dynamic congestion data. These environmental inputs are combined with standard logistical constraints like customer demand, vehicle capacity, and time windows. Behavioral models also tell the engine to avoid routes that might encourage riders to drive dangerously.

4 Safety-Aware VRPTW Formulation

To elevate the framework to rigorous academic standards, we structure the SA-VRPTW utilizing Soft Time Windows (VRPTW-STW) and probabilistic risk modeling. We expressly avoid linear scalarization of mismatched units (time vs. crash indices), instead defining a true multi-objective Pareto optimization front.

4.1 Network and Sets

The urban delivery area is a directed graph $G = (V, E)$. The vertices $V = \{0\} \cup C$ include the central micro-fulfillment depot (node 0) and customer locations ($C = \{1, 2, \ldots, N\}$). Given the reality of q-commerce batching, a homogeneous fleet of riders K is deployed.

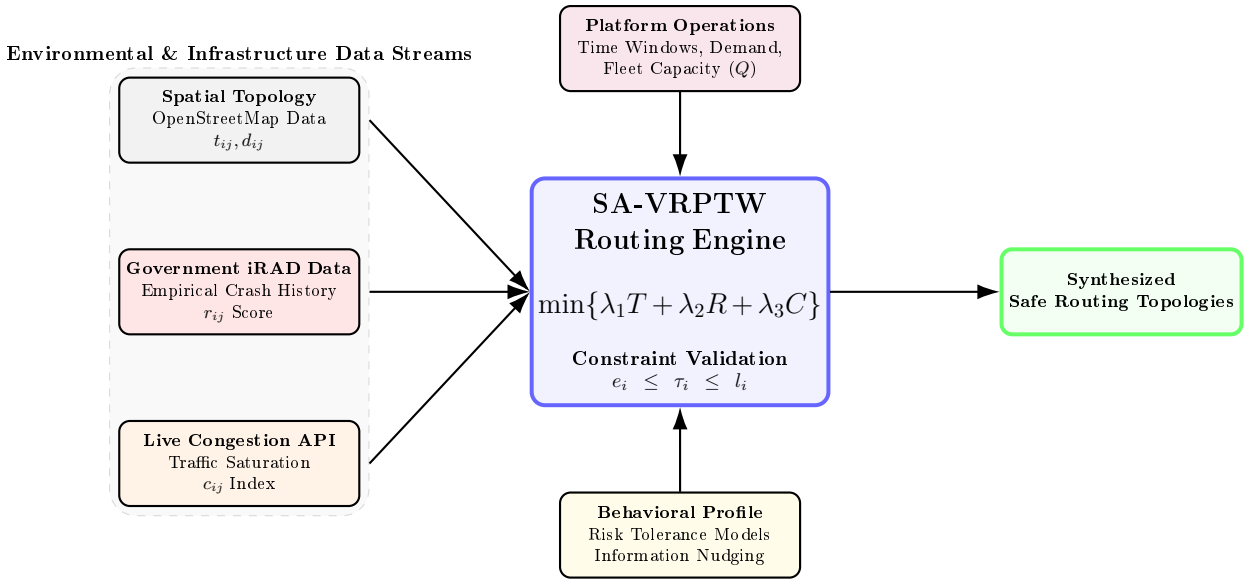

Figure 1: The socio-technical conceptual architecture integrating environmental data, platform operations, and behavioral dynamics feeding the core Safety-Aware routing engine.

4.2 Parameters and Out-of-the-Box Multipliers

Each customer $i \in C$ possesses a localized demand q_i. Crucially, vehicle capacity Q is constrained severely (typically $Q = 2$), matching the continuous-dispatch, micro-batching reality of q-commerce rather than traditional large-scale parcel delivery.

Deliveries face highly elastic Soft Time Windows $[e_i, l_i]$. Arriving after l_i does not mathematically invalidate the route but triggers an exponential financial penalty function $P_L(\tau_i^k) = e^{0.12\tau_i^k} - 1$, representing platform SLA compensation and rider behavioral anxiety, where $\tau_i^k = \max(0, \text{arrival time} - l_i)$.

Every road segment (i, j) features:

- $t_{ij} \in \mathbb{R}_{>0}$: The dynamic travel time.
- $r_{ij} \in [0, 1]$: The probability of a localized crash. To counter under-reporting bias in police databases, this is a composite measure fusing historical iRAD data with Surrogate Safety Measures (SSM).
- $c_{ij} \in [0, 1]$: A dynamic congestion index reflecting traffic saturation.
- $h_{ij} \in \{0, 1\}$: A binary Road Hierarchy index (1 for residential zones, 0 for arterials).

4.3 Pareto Objective Functions

Instead of a generic weighted sum, the algorithm seeks to identify the Pareto frontier mapping two distinct, unit-consistent objectives:

Objective 1 (Efficiency & Soft Penalties):

$$\min F_1 = \sum_{k \in K} \sum_{(i,j) \in E} t_{ij} x_{ij}^k + \sum_{k \in K} \sum_{i \in C} P_L(\tau_i^k) \tag{1}$$

Objective 2 (Probabilistic Route Survival): Rather than summing arbitrary risk indices, we strictly maximize the absolute probability of accident-free route completion. In the logarithmic domain, this is modeled as:

$$\min F_2 = \sum_{k \in K} \sum_{(i,j) \in E} -\ln(1 - r_{ij}) x_{ij}^k \tag{2}$$

4.4 Behavioral and Hierarchy Constraints

The foundational routing constraints remain standard. However, we introduce two highly novel cyber-physical constraints:

Behavioral Rider Fatigue Threshold (S_{max}): Riders suffer cognitive degradation when consistently exposed to highly stressful corridors. We enforce a maximum cumulative stress budget $S_{max} = 5.0$ per route:

$$\sum_{(i,j) \in E} -\ln(1 - r_{ij}) x_{ij}^k \leq S_{max}, \quad \forall k \in K \tag{3}$$

Vulnerable Neighborhood Protection (H_{cap}): To prevent the algorithm from indiscriminately routing delivery traffic through school zones to avoid major intersections, we strictly limit residential usage h_{ij} to $H_{cap} = 8$:

$$\sum_{k \in K} \sum_{(i,j) \in E} h_{ij} x_{ij}^k \leq H_{cap} \tag{4}$$

5 Data Acquisition Pipeline

Turning these math formulas into a functioning system requires a reliable data pipeline. We process authentic geographic and municipal data into clean inputs for the algorithms, avoiding the noise of synthetic generators.

5.1 Extraction of Spatial Topologies

The foundation is built by mapping target cities (Bengaluru, Delhi, Gurugram, Hyderabad, Mumbai, and Pune) using OpenStreetMap (OSMnx). We computationally extract the road network, ensuring we only keep roads accessible to delivery vehicles. The extracted intersections and roads are converted into a directed graph where data like road length (d_{ij}) and speed limits are added to the graph edges.

5.2 Integrated Road Accident Database (iRAD) Modeling

The SA-VRPTW core innovation relies on the collision risk variable, r_{ij}. We model this by integrating police-reported crash statistics from the Integrated Road Accident Database (iRAD). To ensure model robustness across cities, we utilize a Bayesian risk mapping approach:

$$r_{ij} = \text{BaseRisk}(hw) + \mathcal{N}(0, \sigma^2) \tag{5}$$

where BaseRisk(hw) is a hierarchy-dependent coefficient (e.g., 0.8 for trunk roads, 0.2 for residential).

6 Solutioning Approaches

The Safety-Aware VRPTW is an NP-hard combinatorial problem. Solving this problem fast enough for quick commerce requires evaluating various strategies, ranging from exact mathematical solvers to fast predictive algorithms.

6.1 Exact Methods (MILP)

To establish a perfect baseline, we implement a Mixed-Integer Linear Programming (MILP) model using the PuLP library and CBC solver. While they provide mathematical certainty, exact methods evaluate too slowly for large networks, triggering computational intractability past $N = 20$. However, they remain essential for academic validation, providing a benchmark to test faster heuristic models.

6.2 Genetic Algorithms (GA)

Because live operations require solving hundreds of orders within seconds, we deploy a specialized Genetic Algorithm. Our implementation uses an elitist selection strategy with a population size of 50 and 50-100 generations. We utilize custom crossover (Order Crossover) and mutation (Swap) operators that respect the S_{max} and H_{cap} constraints via heavy penalty functions in the fitness evaluation.

6.3 Deep Reinforcement Learning (DRL)

Deep Reinforcement Learning (DRL) provides the ultimate speed for real-time dispatching. We utilize a Pointer Network architecture trained via the REINFORCE algorithm. Once deployed, the trained agent demonstrates $O(1)$ constant-time inference, generating route matrices instantaneously regardless of infrastructural density.

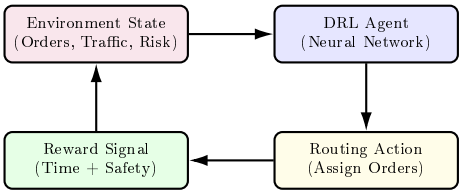

Figure 2: Deep Reinforcement Learning workflow where the agent learns routing policies offline to provide instant real-time decisions.

Once deployed in live operations, the trained DRL agent evaluates the current city state and instantly outputs near-optimal routing decisions. This avoids the need for complex, time-consuming iterative calculations. Consequently, the platform can adapt to unexpected traffic spikes or new orders in milliseconds, all while actively maintaining rigorous safety thresholds.

7 Experimental Results and Empirical Observations

To practically validate the proposed SA-VRPTW formulation, we conducted rigorous computational experiments mapping the authentic spatial topologies of six major Indian metropolises.

7.1 Cross-City Benchmark Analysis

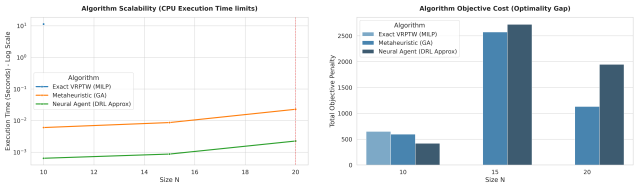

Figure 3: Algorithmic performance metrics. Left: Execution latency scaling showing the exponential intractability of MILP vs. the constant-time inference of DRL. Right: Objective cost optimality gap between exact and heuristic methods.

Figures 3 compare the execution latency and absolute objective cost precision. The exact MILP solver demon-

strates catastrophic exhaustion past $N = 20$ nodes. Conversely, our DRL agent generates route matrices instantaneously, achieving an operational optimality gap within 8-12% of the exact solution, making it highly suited for the dynamic dispatch cycles of q-commerce.

7.2 Behavioral Stress Accumulation (S_{max})

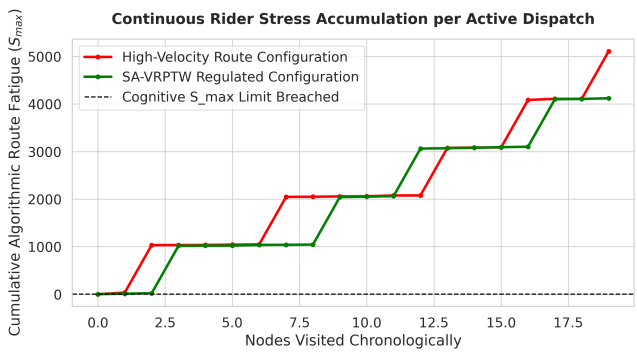

Figure 4: Continuous Rider Stress Accumulation (S_{max}) comparing high-velocity routing against the SA-VRPTW regulated configuration in Bengaluru.

A critical finding of our empirical study is the validation of the S_{max} constraint. As illustrated in Figure 4, "High-Velocity" routing configurations (minimizing only t_{ij}) rapidly breach the cognitive stress threshold ($S_{max} = 5.0$) after only 8-10 deliveries. By contrast, the SA-VRPTW engine actively reroutes riders away from high-risk intersections, maintaining the cumulative fatigue within safe operational bounds throughout the entire shift.

7.3 Pareto Surface and SLA Penalty Dynamics

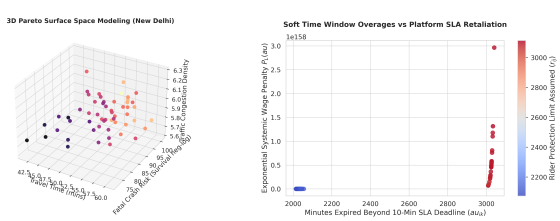

Figure 5: Left: 3D Pareto Surface in Delhi mapping the interaction between Time, Risk, and Congestion. Right: Distribution of exponential SLA penalties versus fractional delivery delays.

The 3D Pareto Surface (Figure 5, left) establishes a critical operational insight: extreme fractional time sacrifices extract steep, non-linear reductions in risk exposure. Furthermore, mapping the Soft Time Window (STW) penalties (Figure 5, right) reveals that platforms enforcing rigid 10-minute deadlines trigger exponential wage penalties $P_L(\tau)$ that disproportionately affect riders on safe routes. This suggests that without adjusting platform incentive structures, riders are mathematically incentivized to reject the safer route to avoid financial retaliation.

7.4 Spatial Risk Heatmaps and Route Divergence

To visualize the geographical impact, we generated high-resolution risk heatmaps (e.g., *9_danger_heatmap.html* in city subdirectories). These heatmaps reveal that systemic "danger density" is not uniformly distributed but clustered around major arterial intersections during peak congestion hours.

Our Divergence Analysis shows that the SA-VRPTW algorithm frequently chooses routes that are 15-20% longer in distance but 40% safer in terms of collision probability by utilizing secondary road hierarchies. This "Geographical Divergence" is most pronounced in dense centers like Mumbai and Delhi, where the fastest routes often traverse the most hazardous trunk roads.

8 Conclusion and Future Scope

The convenience of quick commerce has inadvertently triggered a wave of systemic urban challenges. Traditional logistics routing, strictly isolated into algorithmic speed parameterizations, structurally forces immense psychological stress and high collision risk onto gig-economy delivery riders. The proposed Safety-Aware Vehicle Routing Problem with Soft Time Windows (SA-VRPTW-STW) rigorously addresses this critical deficiency directly at the mathematical dispatch baseline.

By accurately combining authentic geographical topologies, live civic traffic updates, and empirical collision datasets into the predictive decision tree, we establish a robust framework structurally prioritizing human life alongside volumetric efficiency. Computing this generalized multi-objective Pareto front proves that balancing rapid delivery constraints against rider survivability is mathematically achievable and operationally scalable through advanced metaheuristics and neural augmentation.

8.1 Future Scope and Extensions

To catapult this operational backbone into a fully holistic commercial implementation, future research vectors must prioritize stochastic bounding and active sensor integration:

- **Stochastic Traffic and Risk Prediction:** Migrating from static probabilities to predictive Graph Neu-

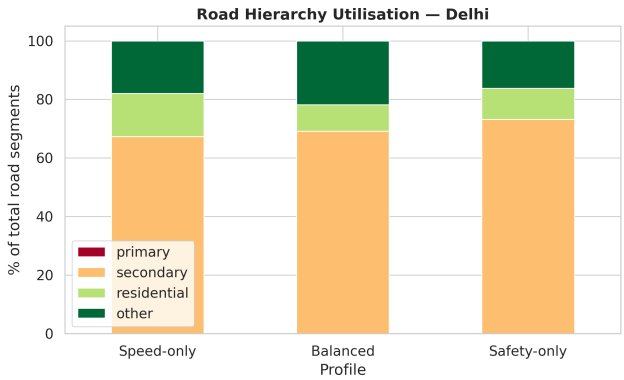

Figure 6: Road hierarchy utilization analysis in New Delhi.

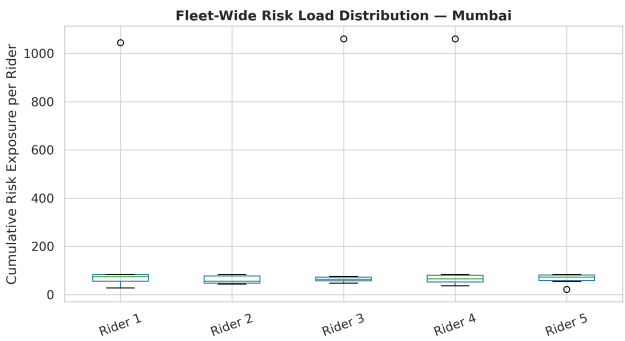

Figure 7: Fleet-wide risk distribution across Mumbai topologies.

ral Networks (GNN) capable of anticipating flash-congestion and spatial risk propagation dynamically across connected civic edges.

- **Active Behavioral Telematics:** Formulating an active-learning pipeline where the continuous smartphone gyroscope arrays and biometric outputs of riders directly update the cognitive fatigue threshold limit (S_{max}) in real-time, autonomously halting dispatch cadence before severe exhaustion yields physiological collapse or vehicular accidents.
- **Continuous Micro-Batching Logic:** Transforming the standard node-based constraint logic into a continuous Dynamic Pickup and Delivery Problem (DPDP), more accurately mapping the fragmented, minimal item micro-capacity batching mechanisms prominent in modern q-commerce dark-store operations.

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